



# Heart Failure

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## Introduction

Heart failure (HF) is a clinical syndrome characterized by different etiologies and pathophysiology. Regardless of the cause, it is generally recognized by a cluster of signs and symptoms including dyspnea, fluid retention, fatigue, and some form of structural or functional heart disease corroborated by either elevated natriuretic peptide levels or objective evidence of cardiogenic pulmonary or systemic congestion. HF is dynamic; symptoms and disease trajectory may change over time and have periods of remission. The most common causes of HF include hypertension, ischemic heart disease and valvular heart disease.<sup>[1]</sup>

HF is associated with significant morbidity and mortality worldwide.<sup>[2]</sup> Its prevalence continues to increase secondary to aging populations with comorbidities and better treatment of acute cardiovascular events. It is associated with significant health-care costs and portends a poor prognosis for patients. Annual mortality rates for HF range between 6–25%, depending on symptoms and hospitalizations.

HF can be categorized in several ways. Most commonly, it is categorized according to left ventricular systolic function by measurement of the left ventricular ejection fraction (LVEF) (see [Table 1](#)). The majority of HF is heart failure with reduced ejection fraction (HFrEF) or heart failure with preserved ejection fraction (HFpEF). Heart failure with mildly reduced ejection fraction (HFmEF) and heart failure with improved ejection fraction (HFimpEF) are newer classifications with evolving clinical implications. It is estimated that about 50% of cases are HFrEF.<sup>[1]</sup>

Table 1: Classification of Heart Failure according to Measured Left Ventricular Ejection Fraction<sup>[3]</sup>

| Classification                                              | Left Ventricular Ejection Fraction (LVEF) Measurement                                    |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Heart failure with reduced ejection fraction (HFrEF)        | HF + LVEF ≤40%                                                                           |
| Heart failure with mildly reduced ejection fraction (HFmEF) | HF + LVEF 41–49%                                                                         |
| Heart failure with preserved ejection fraction (HFpEF)      | HF + LVEF ≥50%                                                                           |
| Heart failure with improved ejection fraction (HFimpEF)     | HF with baseline EF ≤40% + >10 point increase from baseline EF + second measurement >40% |

HFReEF is characterized by decreased pump function (systolic function), dilatation of the left ventricle and decreased LVEF. Most clinical trials have been conducted in patients with HFReEF.

HFpEF is characterized by abnormal myocardial relaxation (diastolic function), nondilated left ventricular with concentric remodelling, or left ventricular hypertrophy and left atrial enlargement and a normal or mildly abnormal LVEF. HFpEF risk factors are older age, female sex/female sex assigned at birth, obesity, hypertension, atrial fibrillation, sleep apnea and diabetes. Diagnosis remains challenging, and treatment is limited but evolving.<sup>[4]</sup> Potentially treatable etiologies include infiltrative, restrictive, inflammatory or genetic cardiomyopathies. These are beyond the scope of this chapter.

HFmEF is poorly classified and may overlap with other categories as a precursor to HFrEF or decompensated HFpEF. More data are required on prognosis and appropriate treatments.<sup>[3][5]</sup>

Data on HFimpEF is also limited. This category attempts to classify patients that have had an improvement or normalization in the LVEF from HFrEF. Some patients in this group may be in “remission” but still have structural abnormalities or may have “recovered” with a treated reversible cause of HF and have no remaining visible structural abnormalities. This is a very heterogeneous group, and treatment strategies will vary depending on the clinical scenario.<sup>[3]</sup>

Table 2: Signs and Symptoms of Heart Failure<sup>[3]</sup>

| Symptoms of HF | Signs of HF                            |
|----------------|----------------------------------------|
| Typical        | More specific                          |
| Bendopnea      | Elevated jugular venous pressure (JVP) |

| Symptoms of HF                     | Signs of HF                          |
|------------------------------------|--------------------------------------|
| Cough                              | Hepatojugular reflux (HJR)           |
| Dyspnea                            | Laterally displaced apical impulse   |
| Exercise intolerance               | Third heart sound (S3)               |
| Fatigue                            | Less specific                        |
| Paroxysmal nocturnal dyspnea (PND) | Cold extremities                     |
| Orthopnea                          | Peripheral edema                     |
|                                    | Pulmonary rales                      |
|                                    | Tachycardia                          |
|                                    | Unintentional weight gain (>2 kg/wk) |

## Goals of Therapy

- Reduce mortality
- Reduce morbidity
  - reduce HF hospitalizations
  - reduce HF exacerbations
  - treat exacerbating factors
  - improve symptoms, exercise tolerance and quality of life
  - prevent disease progression
- Treat modifiable risk factors

## Investigations

The diagnosis of HF includes a detailed clinical history, evaluation of signs and symptoms ([Table 2](#), [Table 3](#)), and initial investigations, including laboratory tests and diagnostic imaging.

**Table 3:** Common Risk Factors for Heart Failure<sup>[6]</sup>

|                        |                                                 |
|------------------------|-------------------------------------------------|
| Age                    | Atrial fibrillation                             |
| Dyslipidemia           | Chemotherapy                                    |
| Hypertension           | Chest radiation                                 |
| Ischemic heart disease | Family history of cardiomyopathy                |
| Obesity                | Substance abuse: alcohol, cocaine, amphetamines |
| Smoking                | Valvular heart disease                          |

**Table 4:** Factors That Can Exacerbate Heart Failure

|                          |                                                      |
|--------------------------|------------------------------------------------------|
| Patient-Specific Factors | Nonadherence to drug therapy or dietary restrictions |
|                          | Anemia                                               |
|                          | Arrhythmias                                          |
|                          | Infections                                           |
|                          | Myocardial ischemia                                  |
|                          | Pulmonary embolism                                   |
|                          | Renal dysfunction                                    |
|                          | Thyroid dysfunction                                  |
|                          | Uncontrolled hypertension                            |
|                          | Valvular heart disease                               |
| Drugs <sup>[7]</sup>     | Drugs that cause sodium and fluid retention:         |
|                          | Androgens                                            |

|                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corticosteroids                                                                                                                                                                                                        |
| Drugs with high sodium content                                                                                                                                                                                         |
| Licorice-containing products                                                                                                                                                                                           |
| Minoxidil                                                                                                                                                                                                              |
| NSAIDs including selective COX-2 inhibitors and high-dose salicylates                                                                                                                                                  |
| Pregabalin                                                                                                                                                                                                             |
| Thiazolidinediones (pioglitazone, rosiglitazone)                                                                                                                                                                       |
| <b>Negative inotropes:</b>                                                                                                                                                                                             |
| Antiarrhythmic agents except amiodarone and dofetilide                                                                                                                                                                 |
| Beta-blockers                                                                                                                                                                                                          |
| Itraconazole                                                                                                                                                                                                           |
| Non-dihydropyridine calcium channel blockers <sup>[8]</sup>                                                                                                                                                            |
| Some anesthesia medications, e.g., propofol                                                                                                                                                                            |
| <b>Cardiotoxic drugs:</b>                                                                                                                                                                                              |
| Alcohol                                                                                                                                                                                                                |
| Amphetamines                                                                                                                                                                                                           |
| Cancer therapies: 5-fluorouracil, alkylating agents (cyclophosphamide, ifosfamide), anthracyclines (doxorubicin, epirubicin, mitoxantrone), bevacizumab, trastuzumab, tyrosine kinase inhibitors (imatinib, sunitinib) |
| Clozapine                                                                                                                                                                                                              |
| Cocaine                                                                                                                                                                                                                |

- Clinical history: symptoms, functional limitations (see [Table 2](#)), history of cardiac disease, risk factors (see [Table 3](#)), exacerbating factors, comorbidities, drugs (see [Table 4](#))
- Physical examination: vital signs, weight, volume status, heart, lung, abdomen, peripheral vascular
- Initial investigations:
  - chest radiograph
  - electrocardiogram

- laboratory tests: natriuretic peptides (see [Table 5](#)), troponin, CBC, electrolytes, serum creatinine/eGFR, urinalysis, glucose, thyroid function tests, iron studies

The use of natriuretic peptide (brain natriuretic peptide [BNP] or N-terminal pro BNP [NT-proBNP]) tests are particularly useful if the diagnosis of HF is unclear as they provide for a “rule in” or “rule out” scenario ([Table 5](#)). In patients with clear signs and symptoms of HF, the use of natriuretic peptide is less useful.

Once a clinical diagnosis of HF is made, echocardiography should be performed in all patients to assess ventricular size and function, including LVEF, and to assess other structures such as cardiac valves (see [Figure 1](#)). Additional testing, including cardiac MRI and cardiac catheterization, may be done to determine specific etiology and select targeted treatment.

**Table 5:** *Natriuretic Peptide Cut-Off Points for the Diagnosis of Heart Failure*<sup>[6]</sup>

|                                                                                                                                 | Cuff-off points (pg/mL) |           |         |           |           |          |
|---------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------|---------|-----------|-----------|----------|
|                                                                                                                                 | BNP                     |           |         | NT-proBNP |           |          |
|                                                                                                                                 | Age <50                 | Age 50–75 | Age >75 | Age <50   | Age 50–75 | Age >75  |
| HF Unlikely                                                                                                                     | <100                    |           |         | <300      |           |          |
| HF Possible                                                                                                                     | 100–400                 |           |         | 300–450   | 300–900   | 300–1800 |
| HF Very Likely                                                                                                                  | >400                    |           |         | >450      | >900      | >1800    |
| <b>Abbreviations:</b> BNP = brain natriuretic peptide; HF = heart failure; NT-proBNP = N-terminal pro brain natriuretic peptide |                         |           |         |           |           |          |

## Therapeutic Choices

Treatment decisions will depend on the underlying etiology of HF, LVEF and symptoms, often described as New York Heart Association Functional Class (see [Table 6](#)). Treatment can include decongestion, acute exacerbation supportive therapy (inotropic support or afterload reduction) and/or chronic maintenance therapy (LV remodelling, neurohormonal blockade, risk factor modification).

### Nonpharmacologic Choices<sup>[6][9][10]</sup>

- Manage concomitant disease states (coronary artery disease and other vascular diseases, diabetes mellitus, dyslipidemia, hypertension, obesity, sleep apnea, supraventricular and ventricular arrhythmias).
- Avoid exacerbating risk factors (see [Table 4](#)).

- Recommend moderate regular physical activity in stable patients. Consider referral to cardiac rehabilitation as required.
- Recommend no more than 1 alcoholic drink per day in all patients. Avoid completely in alcoholic cardiomyopathy.
- Restrict sodium intake in all patients (<2–3 g/day).
- Restrict fluid intake (<1.5–2 L/day) in patients with fluid retention that is not easily controlled with diuretics or in patients with hyponatremia.
- Recommend regular influenza, pneumococcal and SARS-CoV-2 vaccination in all patients.
- Have patients monitor their weight daily. Advise those who gain 0.5 kg/day on several consecutive days or 2 kg in 3 days to alert their team or physician.

**Table 6:** New York Heart Association (NYHA) Functional Classification

| Class | Characteristics                                   |
|-------|---------------------------------------------------|
| I     | No symptoms <sup>[a]</sup> with ordinary activity |
| II    | Symptoms occur with ordinary activity             |
| III   | Symptoms occur with less than ordinary activity   |
| IV    | Symptoms occur at rest or with minimal activity   |

<sup>[a]</sup> Symptoms may include angina, dyspnea, fatigue or palpitations.

### Device Therapy

Given their cost, the following devices should generally be reserved for patients who do not have significant comorbidities that are expected to significantly limit survival.

- Implantable cardioverter defibrillators (ICDs) are recommended for patients with a history of sudden cardiac arrest, ventricular fibrillation (VF) or hemodynamically unstable sustained ventricular tachycardia (VT) in the absence of reversible factors (secondary prevention).
  - Primary ICD placement should be considered in patients with ischemic cardiomyopathy with either:
    - NYHA class II–III and LVEF ≤35%
    - NYHA class I and LVEF ≤30%, where the LVEF is measured at least 1 month post-MI and/or at least 3 months after a coronary revascularization procedure
  - In patients with nonischemic cardiomyopathy, primary ICD therapy should be considered in patients with NYHA class II–III and LVEF ≤35% measured after at least 3 months of optimal medical therapy.

- Cardiac resynchronization therapy (CRT) should be considered in patients with NYHA class II, III and ambulatory NYHA class IV despite optimal medical therapy who are in sinus rhythm with QRS interval ≥130 msec, left bundle branch block (LBBB) QRS morphology and an LVEF ≤35%.<sup>[6][11]</sup> For NYHA class II patients in sinus rhythm receiving optimal medical therapy, CRT is recommended if the QRS duration is ≥130 msec with LBBB QRS morphology and an LVEF ≤30%. Patients with a QRS >150 msec may derive the most benefit from CRT.
- Left ventricular assist devices (LVADs) should be used as a bridge to transplantation, though they are sometimes used as destination therapy for ineligible patients. Use of LVADs as a bridge to recovery is controversial.<sup>[12]</sup>

### Reperfusion and Surgery

- Revascularization, either by percutaneous coronary intervention or coronary artery bypass surgery, should be performed in select patients to improve symptomatic ischemia or dysfunctional myocardium.<sup>[6]</sup>
- Cardiac transplantation should be considered in patients with severe refractory HF, despite optimal therapy, who would otherwise have a good life expectancy.<sup>[6]</sup>

### Pharmacologic Choices

#### Diuretics

Assessment of fluid status in patients with HF is a cornerstone of management. The fluid status of patients with HF is dynamic and variable and should be reassessed regularly. The degree of congestion or fluid overload may vary significantly over time but generally worsens as HF progresses. Regardless of the etiology or classification of HF, the approach to diuresis is the same with the goal being achieving and maintaining euvolemia.

**Diuretics** are used to control signs and symptoms of volume overload. Some patients require only intermittent diuretics and others need chronic therapy. Therefore, the dose of diuretic should be reassessed regularly, with the goal to use the minimum effective dose.<sup>[10]</sup> It is important to highlight that although diuretics are extremely effective in reducing symptoms, they do not modify the natural course of HF and do not decrease mortality. Furthermore, overdiuresis may exacerbate renal dysfunction and/or impact the tolerance of other drugs given to improve HF outcomes (e.g., exacerbate hypotension).

Diuresis in HF usually requires the use of a potent **loop diuretic**, commonly **furosemide**. The dose and route (IV or PO) of diuretic is dependent on several factors including degree of congestion, urgency of decongestion, renal function (lower eGFR requiring higher doses to penetrate the glomeruli), BP, diuretic history and response, and the setting in which it is being administered (inpatient or outpatient). IV formulations may be selected in cases where rapid diuresis is required or in cases where patients may not be responding to oral diuretics (e.g., diuretic resistance or poor absorption due to GI tract edema). Dosing can be daily or multiple times per day, depending on the amount of diuresis required. Doses should be given early in the day to avoid nighttime diuresis and sleep disturbance.

In patients with refractory volume overload, consider the addition of a low-dose **thiazide** intermittently (e.g., a few times/week) to a loop diuretic.<sup>[6][9][10]</sup> The addition of a thiazide counteracts reabsorption of sodium in the distal convoluted tubule, which increases with chronic loop diuretic use. This strategy should be used only by experienced clinicians with close monitoring of weight, renal function and serum potassium because of the risk of severe and potentially fatal dehydration and electrolyte imbalances.<sup>[6][9][10]</sup> **Metolazone** is usually used in this context, but other thiazides, such as hydrochlorothiazide or chlorthalidone, have been tried at doses higher than those used in hypertension.

In terms of monitoring, symptoms can improve within hours or days depending on the route and level of congestion. It is important to monitor both signs and symptoms of congestion and recognize that peripheral edema commonly resolves after pulmonary decongestion. Diuretics (loop and thiazide) are associated with electrolyte abnormalities including hypokalemia, hypomagnesemia and hyponatremia. Serum potassium should be maintained at  $\geq 4$  mmol/L because hypokalemia increases the risk of fatal ventricular arrhythmias and digoxin toxicity.<sup>[6]</sup> Renal function should also be monitored to assess for overdiuresis. It should be noted that poor perfusion from HF can worsen renal function but can subsequently improve with adequate diuresis. Diuretics are also known to increase uric acid levels and may predispose patients to gout; NSAIDs should be avoided if gout treatment is required.

### Agents to Treat Heart Failure with Reduced Ejection Fraction

There has been a recent shift in guidelines, advocating for early initiation of quadruple therapy, including angiotensin receptor/neprilysin inhibitor (ARNI) or ACEI/ARB, beta-blocker, mineralocorticoid receptor antagonist (MRA), and sodium-glucose co-transporter 2 (SGLT2) inhibitor.<sup>[13]</sup> The focus is on starting all 4 classes and then subsequent uptitration to target or maximally tolerated doses, with a goal to optimize in 3–6 months.

There is no preference to order of initiation or uptitration; the approach should be individualized based on patient preference, side effect profile, affordability and comorbidities. There is no proven superior approach to initiation or titration. This strategy is regardless of current HF symptoms and NYHA classification and should be applied to the majority of HFrEF patients (see [Figure 2](#)). Preference should be given to the use of agents for which mortality and morbidity benefits have been established in large, randomized clinical trials (see [Table 7](#)). Drugs used in the treatment of HF are presented in [Table 8](#), and target doses are derived from those achieved in clinical trials.

### ACE Inhibitors

**ACE inhibitors** are recommended in all patients, as they have been shown to improve symptoms and reduce the risk of hospitalization, MI and death in patients with HFrEF.<sup>[14][23][24]</sup> Despite the success of the ARNI valsartan/sacubitril versus enalapril in a large trial, ACE inhibitors are still an initial option for many patients who would not fit the trial entry criteria.<sup>[17]</sup> This includes patients with asymptomatic LV dysfunction, those with severe HFrEF, those with new HFrEF or those who cannot tolerate the equivalent of enalapril 10 mg twice a day, which was used during the trial's run-in period. Reimbursement and affordability also limit the uptake of valsartan/sacubitril.

Start ACE inhibitors at a low dose and titrate at 7- to 14-day intervals to the target dose, or maximum tolerated dose if the target dose cannot be reached. Treatment with target doses is more effective than low doses.<sup>[25]</sup> Measure serum creatinine, potassium and BP before initiating an ACE inhibitor or increasing the dose and 7–14 days after any increase in dosage.<sup>[10]</sup> An increase in serum creatinine of up to 30% is expected and acceptable after initiation of an ACE inhibitor.<sup>[6]</sup> If the increase in serum creatinine is greater than 30%, other causes of worsening renal function should first be excluded. In particular, reassess volume status, as an increase in serum creatinine often arises from hypovolemia due to excessive diuresis. If no other cause for the worsening renal function can be identified, reduce the dose or discontinue the ACE inhibitor. Although data are limited, if the patient's condition improves, periodically reassess the potential of restarting ACE inhibitors or increasing the dose. Patients at greatest risk of hyperkalemia are those with moderate to severe renal dysfunction, high baseline potassium, diabetes mellitus and those receiving potassium-sparing diuretics or MRA.<sup>[26]</sup>

Angioedema remains an uncommon but potentially serious adverse event with ACE inhibitors. A meta-analysis of 26 RCTs reported ACE inhibitor-induced angioedema incidence to be 0.3%.<sup>[27]</sup> In an analysis of a multinational, randomized controlled trial, angioedema was reported more often in the first 3 months of ACE inhibitor use and more frequently in patients self-identifying as Black [OR, 2.88], history of drug rash [OR, 3.78], age >65 years [OR, 1.60] or seasonal allergies [OR, 1.79].<sup>[28]</sup> The association of angioedema with patients who are Black has been found in other trials and, although the mechanism is not established, it is thought to be due to racial differences in sensitivity to bradykinin.

### Angiotensin II Receptor Blockers

Angiotensin II receptor blockers (ARBs) are an alternative to ACE inhibitors in patients with ACE inhibitor-induced cough or angioedema. Patient education and follow-up are essential because ARB-induced angioedema has been reported.<sup>[21]</sup> In patients who are intolerant of ACE inhibitors, **candesartan** has reduced the combined endpoint of cardiovascular death and hospitalizations for HF.<sup>[21]</sup> Data suggest that **valsartan** has a comparable effect on mortality and morbidity to captopril in patients with HF complicating an acute MI.<sup>[29]</sup> Similar to ACE inhibitors, ARBs are associated with renal dysfunction and hyperkalemia; thus, ARBs should not be substituted for ACE inhibitors specifically for these adverse events.<sup>[29]</sup> Considerations for initiation, dosage titration and monitoring for ARBs are the same as for ACE inhibitors.

### Angiotensin Receptor/Neprilysin Inhibitor

Currently, the only **ARNI** available is an equimolar mixture of the ARB **valsartan** and the neprilysin inhibitor **sacubitril**. The neprilysin inhibition leads to the accumulation of bradykinin, adrenomedullin and natriuretic peptides (NPs). The impact on NPs makes the measurement of BNP inappropriate in patients receiving this drug, but the clinical utility of NT-proBNP as a biomarker is not affected. It is important to note that the bioavailability of valsartan is enhanced by sacubitril; valsartan 51 mg in the combination product is approximately equivalent to valsartan 80 mg in single drug formulations.

PARADIGM-HF, a large, multicentre clinical trial that included patients with mild to moderate HF, LVEF <40%, elevated concentrations of NPs or an HF hospitalization in the past 12 months, serum potassium <5.2 mmol/L and an eGFR  $\geq 30$  mL/minute, showed that valsartan/sacubitril was superior to enalapril in reducing the primary endpoint of death from cardiovascular causes or HF hospitalization, both of these endpoints

individually, and death from any cause, as well as improving quality of life.<sup>[17]</sup> Valsartan/sacubitril had a lower risk of hyperkalemia and renal dysfunction than enalapril, but a higher rate of hypotension. Rates of angioedema were similar. Patients in this study needed to be on moderate dose ACEi + beta-blocker +/- an MRA as part of the study entry criteria and were also required to complete a run-in to establish tolerability of both enalapril and valsartan/sacubitril.

Current Canadian HF guidelines recommend the use of valsartan/sacubitril in place of an ACE inhibitor or an ARB in patients with HFrEF who remain symptomatic despite treatment, including during an HF hospitalization. They also recommend that valsartan/sacubitril initiation be considered in patients with newly diagnosed HFrEF admitted to hospital.<sup>[13]</sup>

It is critical to wait 36 hours between the administration of an ACE inhibitor and valsartan/sacubitril (or switching back to an ACE inhibitor) to minimize the risk of angioedema. Following the switch, patients should be educated regarding the risk of hypotension, which is more common than with ACE inhibitors. Valsartan/sacubitril has been studied to a limited extent in patients with a systolic BP <100 mm Hg. Serum potassium and renal function should be monitored. Because valsartan/sacubitril can cause natriuresis, fluid status should be assessed and diuretics adjusted accordingly to avoid volume depletion. Studies have shown that valsartan/sacubitril has a diuretic-sparing effect.<sup>[30]</sup> Dose titration should be performed every 2–4 weeks.

### Beta-Blockers

**Beta-blockers** improve symptoms and reduce the risk of hospitalization and death in patients with HFrEF.<sup>[15][31][32][33]</sup> They are recommended in all patients with an LVEF  $\leq$ 40%.<sup>[13]</sup> Beta-blockers that have been shown to reduce mortality include **bisoprolol**, **carvedilol** and controlled release **metoprolol succinate** (not available in Canada).<sup>[13]</sup> **Metoprolol tartrate** is often used in Canada, extrapolating data from metoprolol succinate, but is generally considered second line to bisoprolol and carvedilol as the data are considered more robust with these agents. In fact, mortality and hospital admissions were significantly more frequent in patients treated with metoprolol tartrate (target dose 50 mg BID) than with carvedilol (25 mg BID) in a large randomized trial.<sup>[34]</sup> This is somewhat controversial as the target dose of metoprolol tartrate used in this trial was lower than the target dose of controlled-release metoprolol succinate that significantly decreased mortality in a large placebo-controlled trial.<sup>[31]</sup> It is hypothesized that perhaps the results were related to inadequate beta-blockade and as a result higher heart rates. Carvedilol may have larger effects on BP secondary to its alpha effects and therefore may be preferred in patients with uncontrolled hypertension.

In HF patients, initiate beta-blockers at a very low dose and slowly titrate the dose at 2- to 4-week intervals.<sup>[6]</sup> Beta-blockers should not be initiated or their doses increased while patients are acutely decompensated, as their negative inotropic effects can worsen HF acutely. Nevertheless, once the patient is clinically stable, beta-blockers should be carefully initiated with close monitoring before the end of the hospitalization in those post acute exacerbation.<sup>[35]</sup> Monitoring should include BP and heart rate before initiating a beta-blocker and before any increase in dose. Additionally, monitor for signs and symptoms of HF decompensation when initiating or increasing the dose of a beta-blocker, paying particular attention to the daily morning weight.<sup>[6]</sup> Patients should be counselled that they may temporarily feel worse when initiating and uptitrating therapy in the short term, but will benefit from long-term reductions in HF outcomes. Beta-blockers should not be stopped abruptly, unless in an emergency situation to avoid rebound tachycardia.<sup>[6]</sup>

### Mineralocorticoid Receptor Antagonists

In patients with HFrEF, an MRA should be used to reduce mortality and morbidity in all patients.<sup>[36][37]</sup>

Mineralocorticoid receptor antagonists (**eplerenone** or **spironolactone**) should not be used in patients with a baseline potassium >5 mmol/L or creatinine clearance <30 mL/minute.<sup>[6]</sup> Monitor vital signs, serum creatinine and potassium at 3 days and 7 days after initiating or titrating the dose of MRAs, and repeat as necessary until the potassium level and renal function are stable.<sup>[6]</sup> Monitoring should then be performed monthly for 3 months and then every 3 months.<sup>[6]</sup>

Eplerenone, unlike spironolactone, does not have anti-androgen effects and therefore does not produce gynecomastia, impotence or irregular menstruation. It is therefore the agent of choice in individuals who have experienced these adverse effects.<sup>[10]</sup> Eplerenone appears to have a similar risk of hyperkalemia and renal dysfunction as spironolactone, so should not be used as a substitute in these situations. Prospective comparative data between these agents is limited. The choice is left to the discretion of the clinician and will likely be influenced by reimbursement considerations.

### Sodium-Glucose Cotransporter 2 Inhibitors

The use of SGLT2 inhibitors such as **dapagliflozin** and **empagliflozin** have been shown to decrease morbidity and mortality, regardless of diabetes status and should be considered in all HFrEF patient unless contraindicated (e.g., type I diabetes mellitus, allergy, history of euglycemic ketoacidosis).<sup>[38][39]</sup> The addition of an SGLT2 inhibitor should especially be prioritized in those with underlying type 2 diabetes mellitus and chronic kidney disease, as there are additive benefits in these populations. Prior to initiation, fluid status should be assessed and diuretic doses adjusted if euvolemic, as SGLT2 inhibitors may cause volume depletion by acting as a mild osmotic diuretic. Some patients may experience worsening renal function, which may be related to volume depletion or secondary to a change in intrarenal hemodynamics (similar to ACE inhibitor or ARB). Clinical trials have consistently shown a “dip” in eGFR in the first 4 weeks, with near resolution by 12 weeks and decline in eGFR slope over time. A drop in eGFR <15% is considered acceptable.

### I<sub>f</sub> Current Inhibitor

**Ivabradine** inhibits the I<sub>f</sub> current in cardiac pacemaker cells resulting in reduced heart rate. It is indicated in Canada for selected individuals with an LVEF  $\leq$ 35% who present with NYHA class II–III symptoms, who are in sinus rhythm with a resting heart rate  $\geq$ 77 bpm despite achieving target or maximally tolerated doses of beta-blockers, or in patients with beta-blocker intolerance. Ivabradine has been shown to reduce the combined risk of CV mortality and HF hospitalizations in a large randomized trial, mainly by reducing the HF hospitalizations.<sup>[20]</sup> Ivabradine also improved symptoms and NYHA functional class. In the trial, the heart rate entry criterion was  $\geq$ 70 bpm, but the reduction in the primary endpoint was seen most in those with a baseline heart rate

≥77 bpm. This heart rate of 77 bpm is used in the Health Canada–approved indications; however, clinical guidelines nonetheless recommend its use in patients with a heart rate ≥70 bpm given the limitations of this subgroup analysis.<sup>[13]</sup> Clinical judgment should be used when considering ivabradine for patients with a heart rate between 70–76 bpm. Ivabradine should not be used in those with atrial fibrillation, as it works on the SA node and will not control heart rate in the setting of atrial fibrillation.

### Digoxin

**Digoxin** improves symptoms and reduces the risk of hospitalization for exacerbations of HFrEF, but does not reduce mortality in patients with persistent moderate to severe symptoms (NYHA class II–IV) while on ACE inhibitor therapy.<sup>[13][22]</sup> Digoxin may be considered in patients with persistent symptoms despite the use of optimal medical therapy (beta-blocker; ACE inhibitor, ARB or ARNI; MRA and SGLT2 inhibitor).<sup>[13]</sup> It may also be considered for ventricular rate control in patients with atrial fibrillation that cannot be controlled by beta-blockers or in patients who cannot tolerate beta-blockers.<sup>[13]</sup> Individualize the dosage based on the patient's age, weight, renal function and concomitant drugs; the usual range is 0.0625–0.25 mg daily. Lower doses may be appropriate in select patients. Given the narrow therapeutic index of digoxin, pay particular attention to identifying and preventing potential drug interactions. Check digoxin serum concentrations when renal function changes significantly, when an interacting drug is added or discontinued, when significant concomitant dehydrating illness occurs, or when digoxin toxicity is suspected. Serum concentrations may also be verified after a steady state is achieved, particularly in older patients and those with renal dysfunction.<sup>[40]</sup> Measure trough serum concentrations at least 8 hours after administration and adjust the dose to maintain the serum concentration between 0.6 and 1.2 nmol/L, which is associated with greater reductions in hospitalization and, possibly, a reduction in all-cause and HF-associated mortality.<sup>[41][42]</sup>

### Nitrates/Hydralazine

The combination of **isosorbide dinitrate** plus **hydralazine** can be used in HF patients who do not tolerate ACE inhibitors, ARBs or an ARNI due to renal dysfunction or hyperkalemia, or as add-on therapy in Black patients.<sup>[6]</sup>

Add-on therapy using isosorbide dinitrate and hydralazine was tested in a randomized controlled trial in self-identified Black patients with NYHA class III–IV HF.<sup>[18]</sup> Patients were already receiving standard therapy (ACE inhibitor, beta-blocker with or without MRA), and add-on therapy showed a reduction in morbidity and mortality.

Nitrate monotherapy is valuable in treating symptoms of angina, paroxysmal nocturnal dyspnea and orthopnea. Nitrates have not been shown to reduce mortality in the absence of hydralazine.<sup>[13]</sup>

### Soluble Guanylate Cyclase Stimulators

**Vericiguat** sensitizes guanylate cyclase to endogenous nitric oxide by stabilizing nitric oxide binding to the binding site. This occurs independent of nitric oxide and is thought to improve smooth muscle relaxation and vasodilation without nitrate tolerance. The multicentre, randomized placebo-controlled VICTORIA trial assessed the efficacy and safety of vericiguat (10mg daily) compared to placebo in patients with a reduced ejection fraction (LVEF < 45%) and chronic HF with recent decompensation.<sup>[43]</sup> The trial results showed a

significant decrease in the primary outcome of first HF hospitalization or cardiovascular death (ARR = 3; NNT = 33 over 10.8 months) when added to contemporary guideline directed medical therapy. There was more symptomatic hypotension and syncope in the vericiguat arm, but there was no statistically significant difference in overall adverse effects. Vericiguat can be considered in high risk patients hospitalized for HF.<sup>[13]</sup>

### Omega-3 Polyunsaturated Fatty Acid Supplementation

The GISSI-HF randomized controlled trial showed a modest reduction of cardiovascular events with low-dose **omega-3 polyunsaturated fatty acid** (n-3 PUFA),<sup>[44]</sup> and some guidelines suggest that low-dose n-3 PUFA (1 g/day) may be considered in patients with mild to moderate HF.<sup>[6]</sup> Doses of more than 3 g/day have been associated with excessive bleeding and may interact with warfarin and DOACs.<sup>[6]</sup> Because n-3 PUFA is a food supplement, patients should consult their pharmacist to help in the selection of a reliable n-3 PUFA brand that most closely matches the formulation studied in HF (850–882 mg eicosapentaenoic acid and docosahexaenoic acid as ethyl esters in an average ratio of 1:1.2). Additionally, one should consider the effect of adding another medication on top of standard HF therapy and its contributions to polypharmacy and adherence, given its modest effect on HF outcomes.

### Calcium Channel Blockers

**Amlodipine** is considered safe, but does not reduce mortality or morbidity in patients with HFrEF.<sup>[45][46]</sup> It may be useful in select patients with persistent angina despite the use of beta-blockers and nitrates or in patients with uncontrolled hypertension despite the use of an ACE inhibitor, MRA, beta-blocker and a diuretic. It should be noted that amlodipine causes dose-related peripheral edema, which may become problematic in those with ongoing fluid retention and overload.

It is generally advised to avoid diltiazem and verapamil in patients with HFrEF because of their negative inotropic effects including adverse outcomes,<sup>[6]</sup> and felodipine and nifedipine because of the lack of data.

### Iron Replacement

Iron deficiency, in the presence or absence of anemia, is common in HF and has been associated with poor HF outcomes such as decreased exercise tolerance, HF hospitalizations and mortality.<sup>[47]</sup> Iron deficiency can present as deficiency in iron stores (absolute deficiency) or a functional deficiency in which iron is not available to be used by cells, such as in states of inflammation like HF. Iron is important to many cell functions including erythropoiesis and other important cell metabolic functions. Iron is especially important to oxygen utilization in cardiac cells. The suggested definition of iron deficiency in HF is serum ferritin <100 ng/mL or serum ferritin 100–299 ng/mL plus transferrin saturation (TSAT) <20%.<sup>[6]</sup>

Guidelines recommend that all patients with HF be assessed for iron deficiency. There is currently no evidence supporting the use of IV iron in HFpEF. In those with HFrEF and iron deficiency (regardless of anemia), consideration should be given to **IV iron replacement** with the aim of improving exercise tolerance, quality of life and reducing HF hospitalizations.<sup>[6]</sup> There is no evidence for use of oral iron supplementation in HF and therefore oral iron should not be used to treat these patients.<sup>[48]</sup>

The use of IV iron has been shown to improve outcomes including exercise tolerance, symptoms of HF, quality of life and heart failure hospitalizations in symptomatic patients with HF<sub>rEF</sub>.<sup>[49]</sup> The majority of larger trials used ferric carboxymaltose; it is unknown if trial results apply to other formulations. Despite this, other IV preparations are commonly used in Canada for this indication and dosed as per individual product monograph recommendations for iron deficiency anemia (see [Common Anemias](#)). The Ganzoni formula is commonly recommended to assess iron deficiency and subsequent replacement (total cumulative dose):<sup>[50]</sup>

iron deficit [mg] = body weight [kg] × (target Hb [g/L] - actual Hb [g/L]) × 0.24 + iron stores [mg]

The IRONMAN trial used intravenous iron (III) isomaltoside in patients with HF and an ejection fraction <45% and found a reduced risk of recurrent HF hospitalizations and cardiovascular death after adjustments due to COVID-19 events.<sup>[51]</sup> This preparation is available in Canada, but does not currently have an indication for iron deficiency in HF.

**Table 7: Outcomes Associated with Drugs Used for Heart Failure with Reduced Ejection Fraction**

| Outcome         | Intervention                                                                                        | Comparator | Relative Risk Reduction (%) | Absolute Risk Reduction (%) | Number Needed to Treat to Prevent 1 Outcome |
|-----------------|-----------------------------------------------------------------------------------------------------|------------|-----------------------------|-----------------------------|---------------------------------------------|
| Death prevented | ACE inhibitor <sup>[14]</sup>                                                                       | Placebo    | 16                          | 4.5                         | 23                                          |
|                 | Beta-blocker <sup>[15]</sup>                                                                        | Placebo    | 32                          | 5.5                         | 19                                          |
|                 | MRA <sup>[16]</sup>                                                                                 | Placebo    | 19                          | 3                           | 33                                          |
|                 | ARNI <sup>[17]</sup>                                                                                | Enalapril  | 16                          | 2.8                         | 36                                          |
|                 | Hydralazine/isosorbide dinitrate (in addition to optimal therapy in Black patients) <sup>[18]</sup> | Placebo    | 39                          | 4                           | 25                                          |
|                 | SGLT2 inhibitor <sup>[19]</sup>                                                                     | Placebo    | 13                          | 1.6                         | 62                                          |
|                 | Ivabradine <sup>[20]</sup>                                                                          | Placebo    | N/A                         | N/A                         | N/A                                         |
|                 | ARB (ACE inhibitor-intolerant patients) <sup>[21]</sup>                                             | Placebo    | 13 <sup>[a]</sup>           | N/A                         | N/A                                         |

| Outcome                 | Intervention                                                                          | Comparator | Relative Risk Reduction (%) | Absolute Risk Reduction (%) | Number Needed to Treat to Prevent 1 Outcome |
|-------------------------|---------------------------------------------------------------------------------------|------------|-----------------------------|-----------------------------|---------------------------------------------|
| Hospitalization avoided | ACE inhibitor <sup>[14]</sup>                                                         | Placebo    | 30                          | 10.8                        | 9                                           |
|                         | Beta-blocker <sup>[15]</sup>                                                          | Placebo    | 33                          | 6                           | 17                                          |
|                         | ARNI (first hospitalization for worsening HF) <sup>[17]</sup>                         | Enalapril  | 21                          | 2.8                         | 36                                          |
|                         | MRA <sup>[16]</sup>                                                                   | Placebo    | 35                          | 6.4                         | 15                                          |
|                         | SGLT2 inhibitor <sup>[19]</sup>                                                       | Placebo    | 31                          | 4.4                         | 22                                          |
|                         | Digoxin <sup>[22]</sup>                                                               | Placebo    | 28                          | 7.9                         | 13                                          |
|                         | Ivabradine <sup>[20]</sup>                                                            | Placebo    | 26                          | 5                           | 20                                          |
|                         | ARB (ACE inhibitor-intolerant patients; hospitalizations for HF only) <sup>[21]</sup> | Placebo    | 32                          | 7.8                         | 13                                          |

<sup>[a]</sup> Study size was not sufficient to produce a statistically significant value in an unadjusted analysis (p = 0.11). This number was significant in an analysis adjusted for covariates (p = 0.033).

**Abbreviations:** ACE = angiotensin-converting enzyme; ARB = angiotensin II receptor blocker; ARNI = angiotensin receptor/neprilysin inhibitor; HF = heart failure; MI = myocardial infarction; MRA = mineralocorticoid receptor antagonist; N/A = not available; SGLT2 = sodium-glucose cotransporter 2

**Antiarrhythmics**

**Amiodarone** may be considered to maintain sinus rhythm in select patients with atrial fibrillation.<sup>[6]</sup> It should be emphasized that a strategy of maintaining sinus rhythm in patients with HF is not superior to ventricular rate control in patients with a history of atrial fibrillation.<sup>[52]</sup>

Amiodarone has also been shown to reduce the frequency of repetitive ICD discharges.

Avoid all other antiarrhythmic drugs.<sup>[6][53]</sup> Antiarrhythmics are not recommended for prevention of sudden cardiac death because they do not reduce mortality.

#### **HMG-CoA Reductase Inhibitors (Statins)**

Although the benefit of **statins** in the primary and secondary prevention of coronary artery disease (CAD) is well established, 2 trials have suggested that the benefits of statins in patients with HF may be limited, even if they have CAD.<sup>[54][55]</sup> Nevertheless, patients included in these trials were relatively older (mean age at baseline of 69 and 73 years) and presented with a baseline LDL cholesterol of only 3.2 mmol/L and 3.5 mmol/L respectively. Consequently, it is reasonable to continue treating younger patients and/or patients who are at high risk of cardiovascular events (recent MI, diabetes and known vascular disease). Because these trials did not study the impact of statin discontinuation, it is reasonable to continue the use of these agents in patients who are receiving them.

#### **Agents to Treat Heart Failure with Preserved Ejection Fraction**

The treatment of HFpEF is evolving. Despite many clinical trials using the same agents found to be beneficial in HFrEF, the majority have been neutral or only showed benefit in subgroups.<sup>[56][57]</sup> The definition and diagnostic criteria for HFpEF have continuously evolved and made it hard to identify and define study populations. Furthermore, it is thought that HFpEF is a heterogeneous group of patients with different phenotypes and differing therapeutic targets.<sup>[58]</sup> It is also hypothesized that those with HFpEF do not have the same neurohormonal activation as HFrEF and perhaps do not respond in the same manner as patients treated with medications used in HFrEF.

Treatment of HFpEF has not been shown to reduce mortality; therefore the focus of treatment has been on control of risk factors (hypertension, diabetes mellitus, ventricular rate in atrial fibrillation) and volume status, improving quality of life, decreasing heart rate to optimize filling time and HF hospitalizations (see [Figure 3](#)).<sup>[6]</sup> However, drug treatment strategies are progressing.

#### **ACE Inhibitor/ARB**

Trials have suggested that the chronic use of renin-angiotensin system modulators does not significantly reduce the risk of mortality in patients with HFpEF.<sup>[59][60][61]</sup> The CHARM-preserved trial showed an improvement in HF hospitalization with **candesartan**, but this was a secondary outcome.<sup>[59]</sup> The study included patients with an LVEF >40%.

#### **Angiotensin Receptor/Neprilysin Inhibitor**

The role of **valsartan/sacubitril** in patients with HFpEF was investigated the PARAGON trial.<sup>[62]</sup> Patients with an LVEF ≥45%, an elevated BNP and evidence of structural heart disease were included. They failed to meet their primary outcome, marginally missing significance. Post-hoc review and analysis by the FDA resulted in an indication for valsartan/sacubitril regardless of ejection fraction, as it felt the results were still clinically significant.<sup>[63]</sup> In Canada, valsartan/sacubitril was not approved for HFpEF based on the PARAGON trial. It can be considered in those with HFpEF to reduce heart failure hospitalizations, acknowledging its limited

evidence. However, given its lack of indication in Canada, reimbursement issues will likely be a barrier to treatment.

#### **Mineralocorticoid Receptor Antagonists**

The TOP-CAT trial was a single trial that investigated the use of **spironolactone** in HFpEF with a LVEF ≥45%.<sup>[64]</sup> This trial failed to meet its primary outcome; however it was the subject of serious concerns around trial data abnormalities. In a post-hoc analysis that removed the questionable sites, the primary outcome was significant. As such, the Canadian HF guidelines now recommend that the use of an MRA such as spironolactone should be considered in individuals with symptomatic HFpEF, serum potassium <5 mmol/L and eGFR ≥ 30 mL/min, with close surveillance of potassium and creatinine.<sup>[6]</sup>

#### **SGLT2 Inhibitor**

There are 2 trials that have investigated the role of SGLT2 inhibitors (**dapagliflozin** and **empagliflozin**) in ambulatory HFpEF patients.<sup>[65][66]</sup> Both studies included those with an LVEF >40% and showed improvement in their primary endpoint of HF hospitalization or CV death, but the outcome was mainly driven by HF hospitalization (ARR = 3.3%; NNT = 30 ).<sup>[65][66][67]</sup> Currently in Canada, empagliflozin has an indication for all HF including HFpEF, and the dapagliflozin HFpEF indication is pending. The SGLT2 inhibitors are the only class of medications that have definitively shown improvement in HFpEF in randomized controlled trials.

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## Choices during Pregnancy and Breastfeeding

### Heart Failure and Pregnancy

Pregnancy is accompanied by important cardiovascular hemodynamic alterations, including increases in blood volume, heart rate and cardiac output, as well as a reduction in peripheral vascular resistance and BP.<sup>[68]</sup> In individuals with existing HF, pregnancy may be associated with significant risk for the patient, depending on the severity of the HF and comorbid conditions.<sup>[68]</sup> This risk is further increased in the presence of other cardiovascular pathologies.<sup>[6]</sup> Because many of the pharmacologic agents used in HF are potentially teratogenic, pharmacologic treatments should be selected and optimized based on their cardiovascular benefits for the patient and risks to the fetus.<sup>[6]</sup>

Peripartum cardiomyopathy (PPCM), defined as HF with LVEF <45% in the last month of pregnancy, or within 5 months of the delivery in the absence of pre-existing cardiac disease,<sup>[6]</sup> has an incidence of 1 in 3000–4000 live births.<sup>[68]</sup>

### Management during Pregnancy

The 2017 Canadian Cardiovascular Society guidelines recommend that patients with pre-existing HF be evaluated and counselled by an expert in HF and pregnancy before becoming pregnant.<sup>[6]</sup> The risk of the pregnancy and of the transmission of inheritable cardiac diseases, if applicable, should be addressed.<sup>[6]</sup>

Patients should be monitored closely during the pregnancy, at the time of delivery and in the postpartum period.<sup>[6]</sup> Should worsening of HF be suspected in a pregnant patient, cardiac testing should include an echocardiogram.<sup>[6]</sup> Refer to the 2017 Canadian Cardiovascular Society guidelines for a more complete discussion.<sup>[6]</sup>

Patients with PPCM whose LVEF does not recover should be advised against any future pregnancy because of the high risk of cardiac mortality and morbidity.<sup>[6]</sup> Inform patients who do recover about the significant risk of recurrence of LV dysfunction in subsequent pregnancies.<sup>[6]</sup>

Sodium and fluid restrictions are recommended to minimize signs and symptoms of hypervolemia. Medications should be switched during pregnancy to those with better safety data.<sup>[6]</sup> Avoid ivabradine and SGLT2 inhibitors during pregnancy because data is limited, and avoid ACE inhibitors, ARBs, ARNI and MRAs because they are known teratogens.<sup>[6]</sup> **Beta-blockers** can be used during pregnancy.<sup>[6]</sup> Beta<sub>1</sub> selective agents, such as **metoprolol**, are generally preferred to avoid increased uterine tone and decreased uterine perfusion.<sup>[6]</sup> Adverse events, such as bradycardia, hypoglycemia and reduced birth weight, have been observed following in utero exposure to beta-blockers. If use of a beta-blocker is needed in the pregnant patient, fetal growth should be monitored during pregnancy and the newborn should be monitored for 48 hours after delivery for bradycardia, hypoglycemia and respiratory depression.<sup>[69]</sup> **Furosemide** can be used to maintain euolemia in the patient, but take care to minimize the risk of dehydration, which can lead to reduced placental perfusion.<sup>[6]</sup> **Digoxin** can also be used safely, but higher doses may be required in pregnant patients because of the increased volume of distribution.<sup>[6]</sup> Reduce the dose of digoxin after delivery. **Hydralazine** in combination with **nitrates** can also be used to manage symptoms.<sup>[6]</sup> Hydralazine is a treatment option in select patients with high BP.<sup>[6]</sup>

In patients with HF prior to pregnancy, maintain chronic treatment (or restart ACE inhibitors) following the pregnancy and continue indefinitely.

In patients with PPCM, pharmacologic therapy should be continued and LV function and size periodically reassessed. In patients in whom ventricular function does not normalize, continue treatment indefinitely. Although no formal guidelines exist, in patients in whom the LV size and function normalize, diuretics can be stopped and beta-blockers and ACE inhibitors can be slowly discontinued over months, or even years, with close monitoring to ensure that the ventricular function does not deteriorate.

### Heart Failure and Breastfeeding

Beta-blockers appear safe in breastfeeding.<sup>[70]</sup> Of the beta-blockers used in HF, **metoprolol** has been the most extensively studied in breastfeeding and may be preferred. Limited data suggest that **furosemide** can be used in breastfeeding.<sup>[70]</sup> The transfer of **spironolactone** into breast milk appears minimal; hence, it can be used during breastfeeding.<sup>[70]</sup> Current data for **enalapril**, **fosinopril**, **quinapril** and **ramipril** indicate that the quantity of these ACE inhibitors present in the breast milk is limited and they can be safely used while breastfeeding.<sup>[70]</sup> Data regarding ARBs and ARNI are limited; therefore, ACE inhibitors are preferred. Ivabradine and SGLT2 inhibitors should be avoided during breastfeeding because it is not known if they are transferred into breast milk.

A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#) and [Drug Use during Breastfeeding](#). Other specialized reference sources are

also provided in these appendices.

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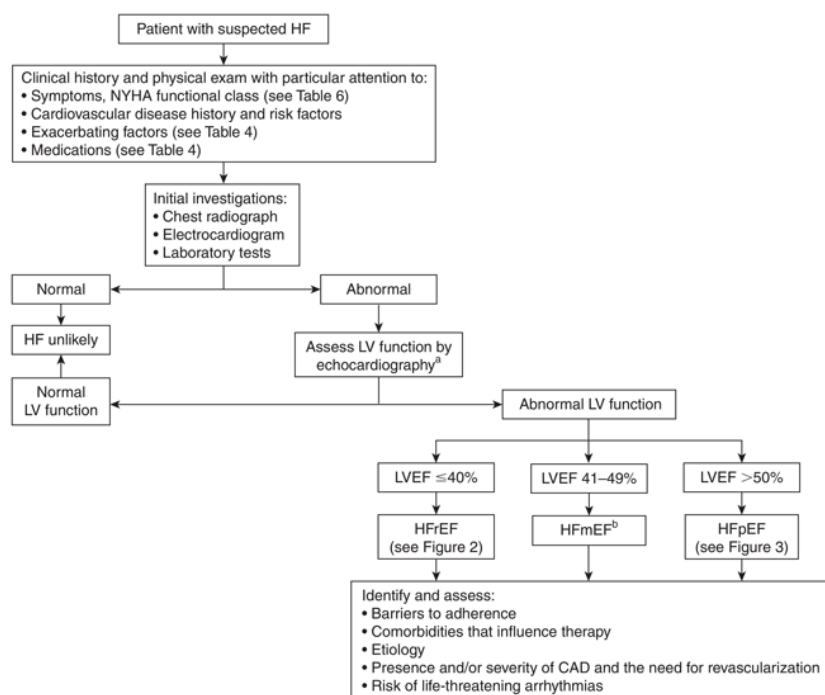
## Therapeutic Tips

- The approach to diuresis is the same regardless of the etiology or type of heart failure. Fluid assessments should occur with every patient visit.
- The goal of diuretics is to maintain the minimum effective dose. Re-assessment should occur at each encounter and especially when initiating SGLT2 inhibitor and ARNI therapy.
- Hypotension or worsening renal function when initiating or increasing the dose of an ACE inhibitor, ARB, ARNI or SGLT2 inhibitor usually indicates the need to reduce the dose of a diuretic.
- Renal dysfunction can be a sign of congestion with HF. Adequate diuresis will improve renal function. Higher doses of diuretics are often required in renal dysfunction to get into the glomeruli and site of action.
- The aim for HFrEF is quadruple therapy. There is no prescribed order for initiation; it is flexible depending on individual patients' parameters. The goal is to initiate and achieve target or maximally tolerated doses within 3–6 months.
- If introduced sequentially, it is not necessary to reach the target doses of one agent before starting the other.
- **Ivabradine** should not be started in patients in whom the beta-blocker has not been optimized. Maximize the beta-blocker dose prior to initiating ivabradine. Ivabradine should not be used in those with or who develop atrial fibrillation.
- In patients with hypotension, consider administering antihypertensive drugs at different times during the day (i.e., staggering doses), rather than at the same time.
- Cough can be a symptom of HF decompensation. Careful evaluation is necessary when evaluating a cough in a patient receiving an ACE inhibitor.
- Electrolyte abnormalities are common in patients with HF. Monitor patients closely, especially when taking combinations of medications that have additive effects, e.g., increased serum potassium due to the combination of an ACE inhibitor, ARB, ARNI plus spironolactone. Measure serum creatinine and electrolytes 7–14 days after the initiation of ACE inhibitors, ARBs, ARNIs, MRAs or diuretics, or following dosage changes.
- An ACE inhibitor, ARB, ARNI, MRAs and SGLT2 inhibitor should all be held as part of sick days management with acute dehydrating illness.<sup>[71]</sup>

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## Algorithms

*Figure 1: Evaluation of Patients with Heart Failure*

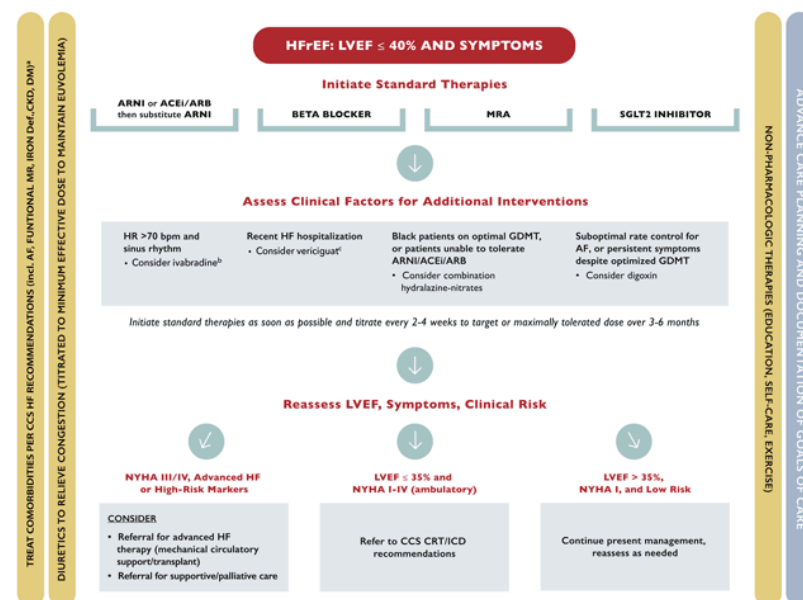


<sup>[a]</sup> Echocardiography (2-dimensional or transthoracic) is used to determine LVEF, chamber size, wall thickness, valve function and presence of pericardial disease. Consider other diagnostic investigations if inconclusive, e.g., magnetic resonance imaging or cardiac catheterization.

<sup>[b]</sup> Data relating to treatment of HFmEF is limited and largely extracted for HFrEF and HFpEF trials. There is reasonable data to support the use of ACEi/ARB, beta-blocker, MRA and SGLT2i in these patients.

**Abbreviations:** CAD = coronary artery disease; HF = heart failure; HFmEF = heart failure with mildly reduced ejection fraction; HFpEF = heart failure with preserved ejection fraction; HFrEF = heart failure with reduced ejection fraction; LV = left ventricular; LVEF = LV ejection fraction; NYHA = New York Heart Association

**Figure 2:** Recommended Management of Heart Failure with Reduced Ejection Fraction<sup>[13]</sup>



<sup>[a]</sup> Information on treating some of these comorbidities can be found in the following chapters: [Chronic Kidney Disease](#),

[Diabetes Mellitus](#), [Supraventricular Tachycardia](#).

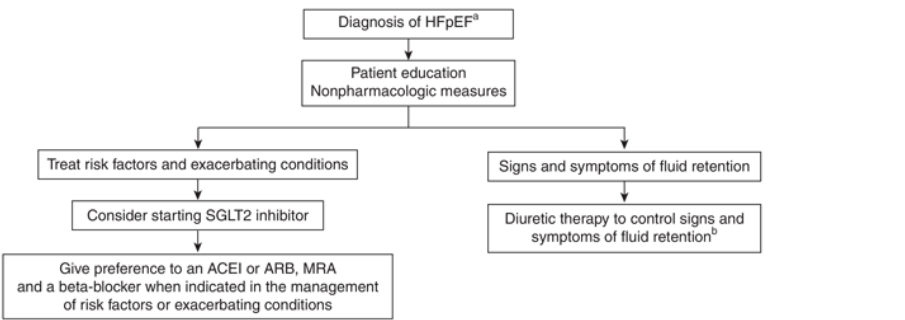
<sup>[b]</sup> Health Canada has approved ivabradine for patients with HFrEF and HR 77 bpm in sinus rhythm.

<sup>[c]</sup> Vericiguat is approved by Health Canada.

**Abbreviations:** AF = atrial fibrillation; ACEi = angiotensin-converting enzyme inhibitor; ARB = angiotensin II receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; CCS = Canadian Cardiovascular Society; CKD = chronic kidney disease; DM = diabetes mellitus; GDMT = goal-directed medical therapy; HF = heart failure; HFrEF = heart failure with reduced ejection fraction; ICD = implantable cardioverter defibrillator; LVEF = left ventricular ejection fraction; MRA = mineralocorticoid receptor antagonist; NYHA = New York Heart Association; SGLT2 = sodium-glucose cotransporter 2

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**Figure 3:** Management of Heart Failure with Preserved Ejection Fraction



<sup>[a]</sup> There is no strong consensus on the definition of HFpEF or its treatment.

<sup>[b]</sup> Diuretics should be used with care because excessive diuresis may decrease cardiac output and compromise renal function in HFpEF.

**Abbreviations:** ACEI = angiotensin-converting enzyme inhibitor; ARB = angiotensin II receptor blocker; HFpEF = heart failure with preserved ejection fraction; MRA = mineralocorticoid receptor antagonist; SGLT2 = sodium-glucose cotransporter 2

Drug Table

Table 8: Drugs Used for Heart Failure

| Drug/Cost <sup>[a]</sup>                                                                                                         | Dosage                      | Adverse Effects                                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                       | Comments                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drug Class: Angiotensin-Converting Enzyme Inhibitors <sup>[b]</sup>                                                              |                             |                                                                                                                                                                            |                                                                                                                                                                                                                                         |                                                                                                                                                                                         |
| <a href="#">enalapril</a> <br>Vasotec, generics | Initial: 1.25–2.5 mg BID PO | Hypotension, hyperkalemia, dry cough, renal insufficiency, angioedema (rare), skin rashes, taste disturbance, proteinuria (rare), neutropenia (rare), headache, dizziness. | Diuretics: hypotension (monitor BP). Potassium-sparing diuretics, potassium supplements, <a href="#">ARBs</a> : hyperkalemia (monitor K <sup>+</sup> ). NSAID: reduced hypotensive effect (monitor BP), fluid retention, renal failure. | Monitor serum creatinine and potassium 7–14 days after initiation of therapy or dose changes. Some experienced clinicians use doses higher than usual target doses. Avoid in pregnancy. |
| < \$25                                                                                                                           | Target: 10 mg BID           |                                                                                                                                                                            |                                                                                                                                                                                                                                         |                                                                                                                                                                                         |

| Drug/Cost <sup>[a]</sup>                                                                                                            | Dosage                          | Adverse Effects                                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                       | Comments                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                     |                                 |                                                                                                                                                                            | Lithium: lithium toxicity (monitor lithium levels).                                                                                                                                                                                     |                                                                                                                                                                                         |
| <a href="#">lisinopril</a> <br>Zestril, generics | Initial: 2.5–5 mg once daily PO | Hypotension, hyperkalemia, dry cough, renal insufficiency, angioedema (rare), skin rashes, taste disturbance, proteinuria (rare), neutropenia (rare), headache, dizziness. | Diuretics: hypotension (monitor BP). Potassium-sparing diuretics, potassium supplements, <a href="#">ARBs</a> : hyperkalemia (monitor K <sup>+</sup> ). NSAID: reduced hypotensive effect (monitor BP), fluid retention, renal failure. | Monitor serum creatinine and potassium 7–14 days after initiation of therapy or dose changes. Some experienced clinicians use doses higher than usual target doses. Avoid in pregnancy. |
| < \$25                                                                                                                              | Target: 20–35 mg once daily     |                                                                                                                                                                            |                                                                                                                                                                                                                                         |                                                                                                                                                                                         |
| <a href="#">perindopril</a><br>Coversyl, generics                                                                                   | Initial: 2 mg once daily PO     | Hypotension, hyperkalemia, dry cough, renal insufficiency, angioedema (rare), skin rashes, taste disturbance, proteinuria (rare), neutropenia (rare), headache, dizziness. | Diuretics: hypotension (monitor BP). Potassium-sparing diuretics, potassium supplements, <a href="#">ARBs</a> : hyperkalemia (monitor K <sup>+</sup> ). NSAID: reduced hypotensive effect (monitor BP), fluid retention, renal failure. | Monitor serum creatinine and potassium 7–14 days after initiation of therapy or dose changes. Some experienced clinicians use doses higher than usual target doses. Avoid in pregnancy. |
| < \$25                                                                                                                              | Target: 8 mg once daily         |                                                                                                                                                                            |                                                                                                                                                                                                                                         |                                                                                                                                                                                         |

| Drug/Cost <sup>[a]</sup>                                                                                                                         | Dosage                                                     | Adverse Effects                                                                                                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                            | Comments                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                  |                                                            |                                                                                                                                                                            | Lithium: lithium toxicity (monitor lithium levels).                                                                                                                                                                                                                                                          |                                                                                                                                                                                               |
| <a href="#">ramipril</a> <br>Altace, generics<br><br>< \$25     | Initial: 1.25–2.5 mg BID PO<br><br>Target: 5 mg BID        | Hypotension, hyperkalemia, dry cough, renal insufficiency, angioedema (rare), skin rashes, taste disturbance, proteinuria (rare), neutropenia (rare), headache, dizziness. | Diuretics: hypotension (monitor BP).<br>Potassium-sparing diuretics, potassium supplements, <a href="#">ARBs</a> : hyperkalemia (monitor K <sup>+</sup> ).<br><br>NSAID: reduced hypotensive effect (monitor BP), fluid retention, renal failure.<br><br>Lithium: lithium toxicity (monitor lithium levels). | Monitor serum creatinine and potassium 7–14 days after initiation of therapy or dose changes.<br>Some experienced clinicians use doses higher than usual target doses.<br>Avoid in pregnancy. |
| <a href="#">trandolapril</a> <br>Mavik, generics<br><br>\$25–50 | Initial: 1 mg once daily PO<br><br>Target: 4 mg once daily | Hypotension, hyperkalemia, dry cough, renal insufficiency, angioedema (rare), skin rashes, taste disturbance, proteinuria (rare), neutropenia (rare), headache, dizziness. | Diuretics: hypotension (monitor BP).<br>Potassium-sparing diuretics, potassium supplements, <a href="#">ARBs</a> : hyperkalemia (monitor K <sup>+</sup> ).<br><br>NSAID: reduced hypotensive effect (monitor BP), fluid retention, renal failure.                                                            | Monitor serum creatinine and potassium 7–14 days after initiation of therapy or dose changes.<br>Some experienced clinicians use doses higher than usual target doses.<br>Avoid in pregnancy. |

| Drug/Cost <sup>[a]</sup>                                                        | Dosage                                                          | Adverse Effects                                                                                                                                 | Drug Interactions                                                                                                                                                                                                                                                                                                                                           | Comments                                                                                                             |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
|                                                                                 |                                                                 |                                                                                                                                                 | Lithium: lithium toxicity (monitor lithium levels).                                                                                                                                                                                                                                                                                                         |                                                                                                                      |
| Drug Class: Angiotensin II Receptor Antagonists                                 |                                                                 |                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                      |
| <a href="#">candesartan</a><br><a href="#">Atacand</a> , generics<br><br>< \$25 | Initial: 4 mg once daily PO<br><br>Target: 32 mg once daily     | Hypotension, hyperkalemia, renal insufficiency, angioedema (rare, less frequent than with <a href="#">ACE</a> inhibitors), headache, dizziness. | Diuretics: hypotension (monitor BP).<br>Potassium-sparing diuretics and <a href="#">ACE</a> inhibitors: hyperkalemia (monitor K <sup>+</sup> ).<br><br>Potassium: hyperkalemia (monitor K <sup>+</sup> ).<br><br>NSAID: reduced hypotensive effect (monitor BP), fluid retention, renal failure.<br><br>Lithium: lithium toxicity (monitor lithium levels). | Monitor serum creatinine and potassium 7–14 days after initiation of therapy or dose changes.<br>Avoid in pregnancy. |
| <a href="#">losartan</a><br>Cozaar, generics<br><br>< \$25                      | Initial: 12.5 mg once daily PO<br><br>Target: 150 mg once daily | Hypotension, hyperkalemia, renal insufficiency, angioedema (rare, less frequent than with <a href="#">ACE</a> inhibitors), headache, dizziness. | Diuretics: hypotension (monitor BP).<br>Potassium-sparing diuretics and <a href="#">ACE</a> inhibitors: hyperkalemia (monitor K <sup>+</sup> ).<br><br>Potassium: hyperkalemia (monitor K <sup>+</sup> ).                                                                                                                                                   | Monitor serum creatinine and potassium 7–14 days after initiation of therapy or dose changes.<br>Avoid in pregnancy. |

| Drug/Cost <sup>[a]</sup>                                   | Dosage                                          | Adverse Effects                                                                                                                                 | Drug Interactions                                                                                                                                                                                                                                                                                                                                           | Comments                                                                                                             |
|------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
|                                                            |                                                 |                                                                                                                                                 | NSAID: reduced hypotensive effect (monitor BP), fluid retention, renal failure.<br><br>Lithium: lithium toxicity (monitor lithium levels).                                                                                                                                                                                                                  |                                                                                                                      |
| <b>valsartan</b><br><b>Diovan</b> , generics<br><br>< \$25 | Initial: 40 mg BID PO<br><br>Target: 160 mg BID | Hypotension, hyperkalemia, renal insufficiency, angioedema (rare, less frequent than with <a href="#">ACE</a> inhibitors), headache, dizziness. | Diuretics: hypotension (monitor BP).<br>Potassium-sparing diuretics and <a href="#">ACE</a> inhibitors: hyperkalemia (monitor K <sup>+</sup> ).<br><br>Potassium: hyperkalemia (monitor K <sup>+</sup> ).<br><br>NSAID: reduced hypotensive effect (monitor BP), fluid retention, renal failure.<br><br>Lithium: lithium toxicity (monitor lithium levels). | Monitor serum creatinine and potassium 7–14 days after initiation of therapy or dose changes.<br>Avoid in pregnancy. |

Drug Class: Angiotensin Receptor–Neprilysin Inhibitor Combinations

|                                                                                                                                                    |                                                                                                |                                                 |                                                                                                                    |                                                                                                                                                                  |
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| <b>valsartan/sacubitril</b><br><br><b>Entresto</b><br><br>~\$235 | Initial: 51/49 mg BID PO<br><br>Target: 103/97 mg BID<br><br>A lower starting dose of 26/24 mg | Hypotension, hyperkalemia, renal insufficiency. | Diuretics: hypotension (monitor BP).<br>Potassium-sparing diuretics, potassium supplements, <a href="#">MRAs</a> : | Reduce the initial dose by half if risk of hypotension is high.<br>Monitor serum creatinine and potassium 7–14 days after initiation of therapy or dose changes. |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Drug/Cost <sup>[a]</sup> | Dosage                                                                                                                                                                | Adverse Effects | Drug Interactions                                                                                                                                                                         | Comments            |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
|                          | BID can be considered in patients at risk of hypotension or those not already receiving the target dose of an <a href="#">ACE</a> inhibitor or an <a href="#">ARB</a> |                 | hyperkalemia (monitor K <sup>+</sup> ).<br><br>NSAID: reduced hypotensive effect (monitor BP), fluid retention, renal failure.<br><br>Lithium: lithium toxicity (monitor lithium levels). | Avoid in pregnancy. |

Drug Class: Beta<sub>1</sub>-adrenergic Antagonists

|                                                                                                                                 |                                                                |                                                                                                                                                                                                              |                                                                                                                                                                 |                         |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>bisoprolol</b> <br>generics<br><br>< \$25 | Initial: 1.25 mg once daily PO<br><br>Target: 10 mg once daily | Orthostatic hypotension, worsening HF/fluid retention, bronchospasm (less with beta <sub>1</sub> -selective), dyspnea, bradycardia, malaise, fatigue, asthenia, erectile dysfunction; may mask hypoglycemia. | Amiodarone, digoxin, diltiazem, ivabradine, verapamil: increased bradycardia.<br>Nondihydropyridine calcium channel blockers: additive cardiodepressant effect. | Selective beta-blocker. |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|

| Drug/Cost <sup>[a]</sup>                                                                     | Dosage                                                 | Adverse Effects                                                                                                                                                                                                                                                                                                                | Drug Interactions                                                                                                                                                                                                    | Comments                                                                                                                                                          |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">carvedilol</a><br>Carvedilol, other<br>generics<br><br>< \$25                    | Initial: 3.125 mg<br>BID PO<br><br>Target: 25 mg BID   | Orthostatic<br>hypotension,<br>worsening<br>HF/fluid retention,<br>bronchospasm<br>(less with beta <sub>1</sub> -<br>selective),<br>dyspnea,<br>bradycardia,<br>malaise, fatigue,<br>asthenia, erectile<br>dysfunction; may<br>mask<br>hypoglycemia.<br>More likely to<br>cause orthostatic<br>hypotension than<br>bisoprolol. | Amiodarone,<br>digoxin, diltiazem,<br>ivabradine,<br>verapamil:<br>increased<br>bradycardia.<br>Nondihydropyridine<br>calcium channel<br>blockers: additive<br>cardiodepressant<br>effect.<br><br>CYP2D6 inhibitors. | Nonselective beta-blocker<br>with alpha-blocking activity.<br>Has some vasodilating<br>effects.                                                                   |
| <a href="#">metoprolol</a><br>Apo-Metoprolol,<br>Metoprolol, other<br>generics<br><br>< \$25 | Initial: 6.25mg BID<br>PO<br><br>Target: 100 mg<br>BID | Orthostatic<br>hypotension,<br>worsening<br>HF/fluid retention,<br>bronchospasm<br>(less with beta <sub>1</sub> -<br>selective),<br>dyspnea,<br>bradycardia,<br>malaise, fatigue,<br>asthenia, erectile<br>dysfunction; may<br>mask<br>hypoglycemia.                                                                           | Amiodarone,<br>digoxin, diltiazem,<br>ivabradine,<br>verapamil:<br>increased<br>bradycardia.<br>Nondihydropyridine<br>calcium channel<br>blockers: additive<br>cardiodepressant<br>effect.<br><br>CYP2D6 inhibitors. | Selective beta-blocker.<br>Formulation available in<br>Canada not shown to be<br>effective in HF.<br>Preferred beta-blocker in<br>pregnancy and<br>breastfeeding. |
| Drug Class: I <sub>f</sub> Current Inhibitor                                                 |                                                        |                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                      |                                                                                                                                                                   |

| Drug/Cost <sup>[a]</sup>                                                                                                                        | Dosage                                                                                       | Adverse Effects                                                                                                                                                                                                                          | Drug Interactions                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Comments                                                                                                                                                                                                                                                                  |
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| <a href="#">ivabradine</a><br>Lancora<br><br>\$50–75                                                                                            | 2.5–7.5 mg BID<br>PO<br><br>Start at a lower<br>dose in patients<br>≥75 y                    | Bradycardia, atrial<br>fibrillation, visual<br>effects (blurred<br>vision,<br>phosphenes).                                                                                                                                               | Avoid concomitant<br>use of QT <sub>c</sub> -<br>prolonging drugs.<br>The concomitant<br>use of strong<br>CYP3A4 inhibitors<br>and verapamil and<br>diltiazem<br>(moderate CYP3A4<br>inhibitors that<br>reduce HR) is<br>contraindicated.<br>Consider starting at<br>a lower dose in<br>patients treated<br>with a moderate<br>CYP3A4 inhibitor.<br>The use of heart<br>rate–reducing<br>agents increases<br>the risk of<br>bradycardia and<br>should be<br>monitored. | Avoid in unstable patients,<br>patients with a prolonged<br>QT <sub>c</sub> or patients at risk of<br>bradycardia.<br>Not indicated in those with<br>concurrent atrial fibrillation.<br>Grapefruit juice increases<br>ivabradine concentrations<br>and should be avoided. |
| Drug Class: Inotropic Agents                                                                                                                    |                                                                                              |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                           |
| <a href="#">digoxin</a> <br>Toloxin, generics<br><br>< \$25 | 0.0625–0.25 mg<br>once daily PO<br>(lower doses may<br>be appropriate in<br>select patients) | Anorexia, nausea,<br>vomiting, visual<br>disturbances,<br>fatigue, dizziness,<br>confusion,<br>delirium, cardiac<br>arrhythmia.<br>Noncardiovascular<br>adverse effects do<br>not always<br>precede<br>potentially fatal<br>arrhythmias. | Amiodarone,<br>clarithromycin,<br>cyclosporine,<br>erythromycin,<br>itraconazole,<br>propafenone,<br>quinidine, ritonavir,<br>tetracycline and<br>verapamil increase<br>digoxin serum<br>levels.<br><br>Antacids,<br>cholestyramine,<br>colestipol,<br>neomycin,                                                                                                                                                                                                       | Adjust dose based on<br>patient's age, weight, renal<br>function and concomitant<br>drugs.<br>Measure trough serum<br>concentrations at least 8 h<br>after administration and<br>adjust the dose to maintain<br>the serum concentration<br>between 0.6 and 1.2 nmol/L.    |

| Drug/Cost <sup>[a]</sup>                                                                                                                              | Dosage                                                              | Adverse Effects                                                                                                                                                                                                                              | Drug Interactions                                                                                                                                                                                                                                                                                                                                                              | Comments                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
|                                                                                                                                                       |                                                                     |                                                                                                                                                                                                                                              | rifampin, St. John's wort and sulfasalazine reduce digoxin serum levels.<br><br>Amiodarone, beta-blockers, diltiazem and verapamil increase risk of bradycardia.                                                                                                                                                                                                               |                                                                             |
| Drug Class: Loop Diuretics                                                                                                                            |                                                                     |                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                |                                                                             |
| <b>bumetanide</b> <br>Burinex<br><br>\$50–75                         | 1–4 mg/day PO                                                       | Dehydration, hypokalemia, hypocalcemia, nausea, hypotension, azotemia, hypomagnesemia, anorexia, hyperglycemia (less than with thiazides), hyperuricemia, weakness, fatigue, rash, increased total cholesterol, ototoxicity with high doses. | Lithium: lithium toxicity (monitor lithium levels).<br>Digoxin: digoxin toxicity if K <sup>+</sup> depleted (monitor K <sup>+</sup> ).<br><br>Corticosteroids: hypokalemia (monitor K <sup>+</sup> ).<br><br>NSAID: reduced diuretic effect; increased renal toxicity (monitor).<br>Thiazide diuretics: hypokalemia, hypomagnesemia, dehydration, renal dysfunction (monitor). | 1 mg is equivalent to 40 mg furosemide.                                     |
| <b>furosemide</b> <br>Lasix, Lasix Special, generics<br><br>< \$25 | 20–500 mg/day administered daily to BID PO<br><br>For decompensated | Dehydration, hypokalemia, hypocalcemia, nausea, hypotension, azotemia,                                                                                                                                                                       | Lithium: lithium toxicity (monitor lithium levels).<br>Digoxin: digoxin toxicity if K <sup>+</sup>                                                                                                                                                                                                                                                                             | To avoid ototoxicity, rate of IV administration should not exceed 4 mg/min. |

| Drug/Cost <sup>[a]</sup>                                                                                                               | Dosage                                                         | Adverse Effects                                                                                                                                                                                                                              | Drug Interactions                                                                                                                                                                                                                                                                                                                                                              | Comments                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                        | HF: may be administered BID or TID<br><br>20–240 mg BID–QID IV | hypomagnesemia, anorexia, hyperglycemia (less than with thiazides), hyperuricemia, weakness, fatigue, rash, increased total cholesterol, ototoxicity with high doses.                                                                        | depleted (monitor K <sup>+</sup> ).<br><br>Corticosteroids: hypokalemia (monitor K <sup>+</sup> ).<br><br>NSAID: reduced diuretic effect; increased renal toxicity (monitor).<br>Thiazide diuretics: hypokalemia, hypomagnesemia, dehydration, renal dysfunction (monitor).                                                                                                    |                                                                                                                       |
| <b>ethacrynic acid</b> <br>Edecrin<br><br>\$100–125 | 50–200 mg daily or divided BID PO                              | Dehydration, hypokalemia, hypocalcemia, nausea, hypotension, azotemia, hypomagnesemia, anorexia, hyperglycemia (less than with thiazides), hyperuricemia, weakness, fatigue, rash, increased total cholesterol, ototoxicity with high doses. | Lithium: lithium toxicity (monitor lithium levels).<br>Digoxin: digoxin toxicity if K <sup>+</sup> depleted (monitor K <sup>+</sup> ).<br><br>Corticosteroids: hypokalemia (monitor K <sup>+</sup> ).<br><br>NSAID: reduced diuretic effect; increased renal toxicity (monitor).<br>Thiazide diuretics: hypokalemia, hypomagnesemia, dehydration, renal dysfunction (monitor). | An alternative diuretic in patients with sulfonamide hypersensitivity.<br>Higher risk of ototoxicity than furosemide. |
| Drug Class: Mineralocorticoid Receptor Antagonists (MRAs)                                                                              |                                                                |                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                       |

| Drug/Cost <sup>[a]</sup>                                                                                                                     | Dosage                                                                                        | Adverse Effects                                                                            | Drug Interactions                                                                                                                                                                                                                                                                                            | Comments                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>eplerenone</b> <br>Inspra, generics<br><br>\$50–75       | Initial: 25 mg once daily PO or every 2 days<br><br>Target: 50 mg once daily                  | Hyperkalemia, dehydration, dizziness, diarrhea, nausea.                                    | Diuretics: hypotension (monitor BP).<br>Potassium-sparing diuretics, potassium supplements, <a href="#">ARBs</a> : hyperkalemia (monitor K <sup>+</sup> ).<br><br>NSAID: reduced hypotensive effect (monitor BP), fluid retention, renal failure.<br><br>Lithium: lithium toxicity (monitor lithium levels). | Monitor serum creatinine and potassium 3 and 7 days after initiating or titrating dose. Repeat every 1–3 months once stable. Avoid in pregnancy. |
| <b>spironolactone</b> <br>Aldactone, generics<br><br>< \$25 | Initial: 12.5 mg daily or every 2 days PO<br><br>Target (for mortality benefit): 25–50 mg/day | Hyperkalemia, dehydration, nausea, gynecomastia (usually reversible upon discontinuation). | <a href="#">ACE</a> inhibitors, <a href="#">ARBs</a> , potassium supplements: hyperkalemia.<br>NSAID: reduced diuretic effect, worsening renal function, hyperkalemia.                                                                                                                                       | Monitor serum creatinine and potassium 3 and 7 days after initiating or titrating dose. Repeat every 1–3 months once stable. Avoid in pregnancy. |

Drug Class: Sodium-Glucose Cotransporter 2 Inhibitors

|                                                                             |                     |                                                                                                                             |                                                                                                                |                         |
|-----------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>dapagliflozin</b><br><a href="#">Forxiga</a> , generics<br><br>\$100–125 | 10 mg PO once daily | Volume depletion, urinary tract infections, genital mycotic infections. In those with T2DM:<br><br>hypoglycemia, euglycemic | Reassess diuretic dose; consider dose reduction if euvolemic. Reassess sulfonyleurea and insulin dose to avoid | Contraindicated in T1DM |
|-----------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------|

| Drug/Cost <sup>[a]</sup>                                                                                                                               | Dosage              | Adverse Effects                                                                                                                                    | Drug Interactions                                                                                                                               | Comments                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
|                                                                                                                                                        |                     | diabetic ketoacidosis.                                                                                                                             | hypoglycemia in those with T2DM.                                                                                                                |                         |
| <b>empagliflozin</b> <br><a href="#">Jardiance</a><br><br>\$100–125 | 10 mg PO once daily | Volume depletion, urinary tract infections, genital mycotic infections. In those with T2DM:<br><br>hypoglycemia, euglycemic diabetic ketoacidosis. | Reassess diuretic dose; consider dose reduction if euvolemic. Reassess sulfonyleurea and insulin dose to avoid hypoglycemia in those with T2DM. | Contraindicated in T1DM |

Drug Class: Soluble Guanylate Cyclase Stimulators

|                                               |                                                                                                           |                                |                         |                |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------|----------------|
| <b>vericiguat</b><br>Verquvo<br><br>\$100–125 | Initial: 2.5 mg once daily PO<br><br>If tolerated increase dose every 2 wks to target dose of 10 mg daily | Hypotension, anemia, dizziness | Sildenafil: hypotension | Take with food |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------|----------------|

Drug Class: Thiazides and Related Diuretics

|                                                                                                                                    |                                                                                                                |                                                                                                                                                           |                                                                                                                                                                                                    |                                                                                          |
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| <b>metolazone</b> <br>Zaroxolyn<br><br>< \$25 | 2.5–10 mg once daily PO<br><br>(use lower doses in combination with a loop diuretic; 2.5–5 mg daily to weekly) | Dehydration, hypokalemia, nausea, hypotension, azotemia, hypomagnesemia, hypercalcemia, hyponatremia, hyperglycemia (more with thiazides), hyperuricemia, | Lithium: lithium toxicity (monitor lithium levels). Digoxin: digoxin toxicity if K <sup>+</sup> depleted (monitor K <sup>+</sup> ).<br><br>Corticosteroids: hypokalemia (monitor K <sup>+</sup> ). | Experienced clinicians usually combine with a loop diuretic in those with refractory HF. |
|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|

| Drug/Cost <sup>[a]</sup> | Dosage | Adverse Effects                    | Drug Interactions                                                                                                                                             | Comments |
|--------------------------|--------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                          |        | rash, increased total cholesterol. | NSAID: reduced diuretic effect; increased renal toxicity (monitor).<br>Loop diuretics: hypokalemia, hypomagnesemia, dehydration, renal dysfunction (monitor). |          |

Drug Class: Vasodilators

|                                         |                             |                                                        |                                                           |                                               |
|-----------------------------------------|-----------------------------|--------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------|
| <b>hydralazine</b><br>generics          | Initial: 10–25 mg<br>TID PO | Hypotension, GI complaints, <b>SLE</b> -like syndrome. |                                                           | Use in combination with isosorbide dinitrate. |
| \$25–50                                 | Target: 75 mg<br>TID–QID    |                                                        |                                                           |                                               |
| <b>isosorbide dinitrate</b><br>generics | Initial: 10–20 mg<br>TID PO | Headache, hypotension.                                 | Sildenafil, vardenafil and tadalafil: severe hypotension. | Use in combination with hydralazine.          |
| < \$25                                  | Target: 40 mg<br>TID–QID    |                                                        |                                                           |                                               |

<sup>[a]</sup> Cost of 30-day supply for target doses; includes drug cost only.

<sup>[b]</sup> Only captopril, enalapril, ramipril and trandolapril have been shown to reduce morbidity and prolong survival in heart failure or in patients with LV dysfunction post-MI. Captopril is not included in the table because 3x daily dosing is required.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

**Abbreviations:** ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; BP = blood pressure; GI = gastrointestinal; HF = heart failure; HR = heart rate; LV = left ventricular; MI = myocardial infarction; MRA = mineralocorticoid receptor antagonist; NSAIDs = nonsteroidal anti-inflammatory drugs; SLE = systemic lupus erythematosus; T1DM = type 1 diabetes mellitus; T2DM = type 2 diabetes mellitus

## Suggested Readings

Heidenreich PA, Bozkurt B, Aguilar D et al. 2022 American College of Cardiology/American Heart Association/Heart Failure Society of America guideline for the management of heart failure: executive summary. *J Card Fail* 2022;(5):810–30.

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